



Primary Market Daily Report Assumptions

Daily Originator Mortgage Rates

Points

$$\text{Total Points} = \text{Origination Points} + \text{Discount Points}$$

We normalize the rate to 1-point rate:

$$1 \text{ Point Rate} = \text{Base Rate} - \frac{(1\% - \text{Total Points})}{4}$$

We can observe the Base Rate, APR and Discount Points directly from the various bank websites, but we need to infer the origination points (from the APR). The rates are observed daily on 4:30pm.

FHA Rates

Assume that banks include the Upfront Mortgage Insurance Premium (UFMIP) in their closing cost estimates:

$$\text{Origination Points} = \frac{\text{Closing Cost}}{\text{Loan Size}} - \text{UFMIP}$$

$$\text{Discount Points} = 0\%$$

$$\text{Annual Mortgage Insurance Premium} = 85 \text{ bps}$$

Servicing Spread

We generate FNMA 30-year and GNMA 30-year current coupon rates every day based on TBA prices.

For the Conventional, Jumbo Conforming, and Jumbo Non-Conforming sectors:

$$\text{Servicing Spread} = 1 \text{ Point Rate} - \text{FNMA 30 year Current Coupon Rate}$$

For FHA loans:

$$\text{Servicing Spread} = 1 \text{ Point Rate} - \text{GNMA 30 year Current Coupon Rate}$$

Mortgage Loan Assumptions for the Base Rate

All calculations are based on refinance rates.

		Loan Amount	Down Payment
Wells Fargo	Conventional	\$200,000	25%
	FHA	\$200,000	3.50%
	Jumbo	\$500,000	25%
	Jumbo Non-Conforming	\$750,000	25%
Bank of America	Conventional	\$200,000	20%
Chase	Conventional	\$200,000	20%
Quicken	Conventional	\$200,000	30%

Refinance rates are based on the refinance of a single-family, primary residence.

Proxy for Pipeline Hedging Costs

Realized Current Coupon Volatility

Realized Volatility is the annualized standard deviation of daily absolute difference of current coupon. This volatility is calculated in an interval of 60 days.

Swaption Volatility

Here, we use 3m X 5y Swaption Normal Volatility

Servicing Spread and FHLMC Survey Rate

Primary Mortgage Rates

Freddie Mac's Primary Mortgage Market Survey (PMMS) 30-year fixed-rate is used as the weekly primary market rate. The report presents the 1-point PMMS rate using the raw rate and points provided by Freddie Mac. These rates are received every Thursday.

Weekly Current Coupon

The servicing spread is the difference between the primary (PMMS) and secondary mortgage rates. Although the PMMS rate is a weekly weighted average rate based on data from Monday to Wednesday, in order to smooth out the volatility and capture current coupon changes from the preceding Thursday and Friday, in practice we use the weights as below:

$$\text{Weekly Current Coupon} = 20\% \times \text{Monday} + 20\% \times \text{Tuesday} + 20\% \times \text{Wednesday} + 20\% \times \text{Thursday} + 20\% \times \text{Friday}$$

Servicing Spread

Weekly spread between PMMS rate and weighted average current coupon:

$$\text{Servicing Spread} = \text{PMMS} - \text{Weekly Weighted Average Current Coupon}$$

Time from Refinancing Application to Funding

Time-to-Close

The time from loan application to funding.

Data Source and Sampling

We use the Ellie Mae's Origination Insight Report as data source. This report mines its application data from a robust sampling of approximately 66% of all 3.7 million mortgage applications that were initiated on the Encompass origination platform. Only refinance loan data is used in this report. The report is generated monthly, and there is a two month lag between the current month and the last report.

Effective Guarantee Fees and Upfront Loan-Level Price Adjustments

SF Effective g-fee

Fannie Mae single-family effective g-fee rate is produced on Fannie Mae annual and quarter SEC filing documents.

SF New Issued g-fee

Fannie Mae single-family average charged g-fee on new acquisitions.

Upfront Loan-Level Price Adjustments (LLPAs)

Adverse market Delivery Charge (AMDC) of 0.25% has been added to the LLPAs numbers in the matrix by LTV and credit score.

PMI

Data is captured from MGIC Rate Cards and Program Highlights - Monthly Premiums Report. Using data for 30-year Fixed Rate; Coverage Requirements: for loan terms > 20 years.

FHA Insurance Premiums

Mortgage insurance premium is update by FHA Mortgagee Letters.



Primary Market Daily Report Assumptions

Origination Market Concentration

Lag Between Asof Date and Issued Date

Before 6/1/2011, there is one month lag between Asof and Issued date. But after 6/1/2011, there is no lag between them (Asof = Issued date). We use two algorithms to fit both periods and combine them together on 6/1/2011.

Trends in Specified Pool Issuance

Definitions for the Specified Pools:

Non-TBA Eligible Specified Pools

Jumbo	Loan Size $\geq$ 417,000
CQ	LTV $>$ 125
CR	$105 < \text{LTV} \leq 125$

TBA Eligible Specified Pools

Low FICO	FICO $<$ 700
MHA	100% Refi & $80 < \text{OLTV} \leq 105$
Investor	100% Investor
175K	$150\text{k} < \text{AOLS} \leq 175\text{k}$
HLB	$110\text{k} < \text{AOLS} \leq 150\text{k}$
MLB	$85\text{k} < \text{AOLS} \leq 110\text{k}$
LLB	$\text{AOLS} \leq 85\text{k}$
Geo	NY, PR

FHA Single-Family Application Activity

Does not include Home Equity Conversion Mortgage (HECM)

FHA Streamline Low MIP:

MIP decreased on June 11, 2012 for targeted streamline refinance loans to 1/55 bps